

ASHUTOSH PRADHAN

Software Developer · Distributed Systems · AI-Driven Solutions

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[GitHub](#) | [LinkedIn](#)

SUMMARY

Results-driven Software Developer with hands-on experience building scalable, high-performance applications. Focused on deepening expertise in **distributed systems** and **AI-driven solutions**, with the long-term goal of becoming a skilled software architect who designs resilient systems that solve complex real-world problems and create meaningful impact.

WORK EXPERIENCE

FiveOne — Kathmandu, Nepal

Aug 2025 – Present

Software Developer

- Architected a modular **NestJS + MongoDB** backend with a **ports-and-adapters (hexagonal)** design, powering a scalable and maintainable real-time social platform.
- Integrated **REST** and **WebSocket** APIs to synchronize AI events, real-time game states, and chat interactions for seamless multiplayer experiences.
- Implemented AI-powered conversational features using the **OpenAI LLM API** with **prompt engineering**, increasing per-session user engagement by **~25–35%** and reducing early chat drop-offs.
- Integrated **Firebase** (Auth, Analytics, Messaging) with **Google/Apple Sign-In** and **Notifee** for secure login and reliable push-notification engagement.

Gunasys — Kathmandu, Nepal

Apr 2024 – Aug 2025

Software Developer

- Implemented secure authentication and **role-based access control (RBAC)** to manage permissions across multiple business roles and departments.
- Developed and optimized **RESTful** backend services and database schemas, integrating **in-memory caching** to reduce average transaction latency to **<150ms**.
- Improved frontend performance through state-management optimization, caching strategies, and efficient API handling for smoother user experiences.
- Built **LLM-powered RAG pipelines** to deliver a domain-specific medical chatbot supporting one-on-one patient queries.

Islington College — Kathmandu, Nepal

Feb 2025 – Aug 2025

Academic Liaison Officer

- Conducted structured student follow-ups on attendance and coursework submissions, driving a **35% reduction** in non-submissions and a measurable increase in overall attendance.
- Managed academic staff scheduling and workload allocation, achieving **zero class cancellations** throughout tenure and significantly improving timetable efficiency.

Fresh KTM — Kathmandu, Nepal

Oct 2023 – Apr 2024

Frontend Intern

- Designed and deployed user-centric workflows with dynamic dashboards and real-time updates, improving inventory tracking and order-management efficiency by **~25%**.
- Collaborated with senior developers to enhance FreshKTM's **ERP** web app, implementing responsive frontend components and state-management optimizations.

PROJECTS

KaKhani — Restaurant Reviews with NLP | *FastAPI, Next.js, Python, PostgreSQL, Jupyter* kakhani.netlify.app

- Designed and developed a full-stack restaurant review platform enabling users to submit, browse, filter, and rate restaurants with paginated, searchable listings.
- Implemented **RESTful** backend services using **FastAPI**, following a clean layered architecture with DTOs, service abstraction, and repository patterns.
- Built an **NLP**-based review-analysis pipeline that extracts sentiment scores and keywords, feeding aggregate restaurant ratings and sentiment-weighted rankings.
- Integrated **Redis** caching with timed and action-based invalidation, and secured the API with **JWT**-based authentication and role-based authorization.

Anyani — Animated Image Generator | *Python, Next.js, Jupyter Notebook* anyani.netlify.app

- Built an animated face-generation pipeline using a **ResNet-18** feature extractor combined with **Conv2D** layers for facial-attribute encoding.
- Implemented a **GAN** architecture (Generator + Discriminator) in **TensorFlow / PyTorch** to synthesize animated face images.
- Optimized the model with **CUDA**, batch normalization, and residual connections to improve performance and convergence.
- Deployed the generator behind a **FastAPI** backend with a **Next.js** frontend for on-demand animated-face generation.

EDUCATION

Islington College

Oct 2021 – Dec 2024

BSc (Hons) Computer Science and Information Technology — First Class Honours

Rato Bangla School

Jun 2018 – Oct 2020

A Levels — 3.4 GPA

SKILLS

Languages	Java, Python, C#, JavaScript
Frontend	React.js, React Native, Next.js, Context API, Vue
Backend	Node.js, NestJS, FastAPI, Spring Boot, Django, ASP.NET
Databases	PostgreSQL, MongoDB, SQLite
Caching	Redis, in-memory caching
AI / ML	RAG pipelines, LLM integration, prompt engineering, NLP, TensorFlow, PyTorch
Cloud / DevOps	AWS (EC2, S3), Docker, CI/CD (GitHub Actions), Firebase
API / Realtime	REST APIs, GraphQL, WebSockets
Auth & Security	JWT, OAuth, RBAC
Testing	JUnit, PyTest
Architecture	Microservices, Ports-and-Adapters (Hexagonal)
Tools	Git, Postman, Jupyter, IntelliJ, Eclipse, Visual Studio, Android Studio, Anaconda
Methodologies	Agile
OS	Windows, Linux

SOFT SKILLS

Adaptability · Problem Solving · Communication · Management · Teamwork

CERTIFICATIONS

- **AWS Academy Cloud Foundations** — 2022
- **AWS Academy Machine Learning Foundations** — 2022

AWARDS

- **AAA Scholarship**, Islington College — July 2025

REFERENCES

Birat Thapa — Chief Backend Developer, Fresh KTM
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Bishal Wosti — Project Manager & Head, Gunasys
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